

Paper Abstract

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Selected Paper: Performance Implications of Packet Filtering with Linux eBPF

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Authors: Dominik Scholz, Daniel Raumer, Paul Emmerich, Alexander Kurtz, Krzysztof Lesiak, and Georg Carle

Project Title: DefenseNet: Behavioral eBPF Firewall with Real-Time Response

The paper **Performance Implications of Packet Filtering with Linux eBPF** examines how extended Berkeley Packet Filter (eBPF) improves packet filtering efficiency by processing packets at the earliest stage in the Linux networking stack. The authors compare eBPF-based filtering techniques, such as the eXpress Data Path (XDP), against traditional methods like iptables and nftables, demonstrating that eBPF significantly reduces latency and CPU overhead while maintaining flexibility in firewall rule deployment.

Building upon this foundation, this project implements an enhanced eBPF-based firewall that extends beyond static filtering to provide behavioral threat detection and automated response capabilities. The project will contribute in the following ways:

- **Rate-Based Attack Detection:** Per-source IP rate monitoring with automatic detection and mitigation of volumetric DDoS attacks through configurable packets-per-second thresholds.
- **Automated Threat Mitigation:** Dynamic blacklist management that automatically blocks malicious sources based on observed behavior without manual intervention.
- **Enhanced Web-Based Interface:** Real-time dashboard with traffic visualization, attack detection alerts, interactive rule management, and comprehensive analytics.
- **IPv6 Protocol Support:** Extended filtering capabilities for both IPv4 and IPv6 to protect modern dual-stack networks.

By implementing adaptive rate monitoring, automated blocking mechanisms, and comprehensive visibility tools, this project demonstrates how eBPF can evolve from a high-performance static filter into an intelligent, self-defending firewall system. We will evaluate the system against various attack scenarios measuring detection accuracy, false positive rates, throughput impact, and response latency. Additionally, we will explore the potential integration of machine learning techniques for enhanced anomaly detection and adaptive threshold optimization.